

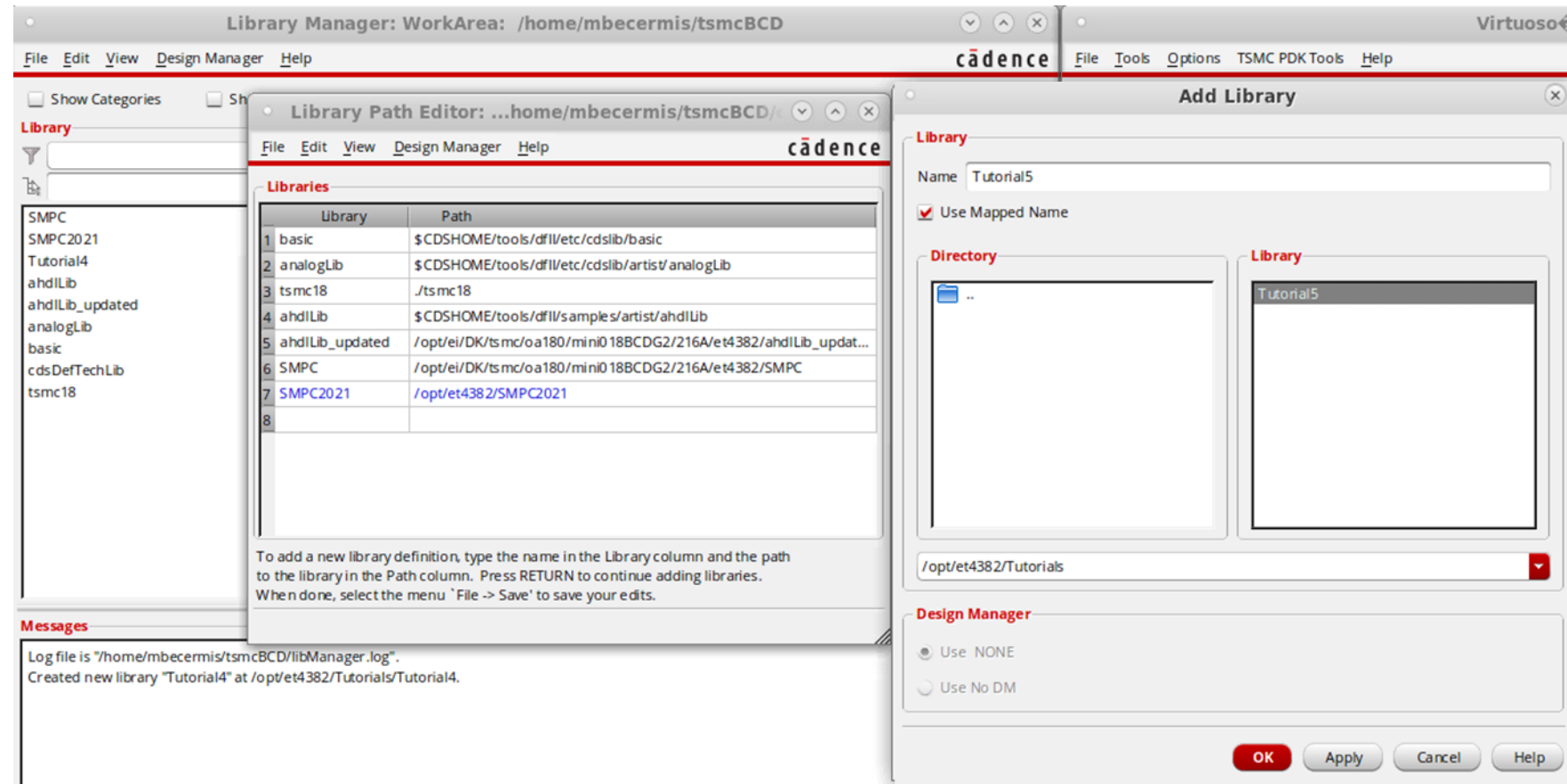
ET4382: Power Conversion Techniques

Tutorial 5

11th March, 2021

Please open Virtuoso

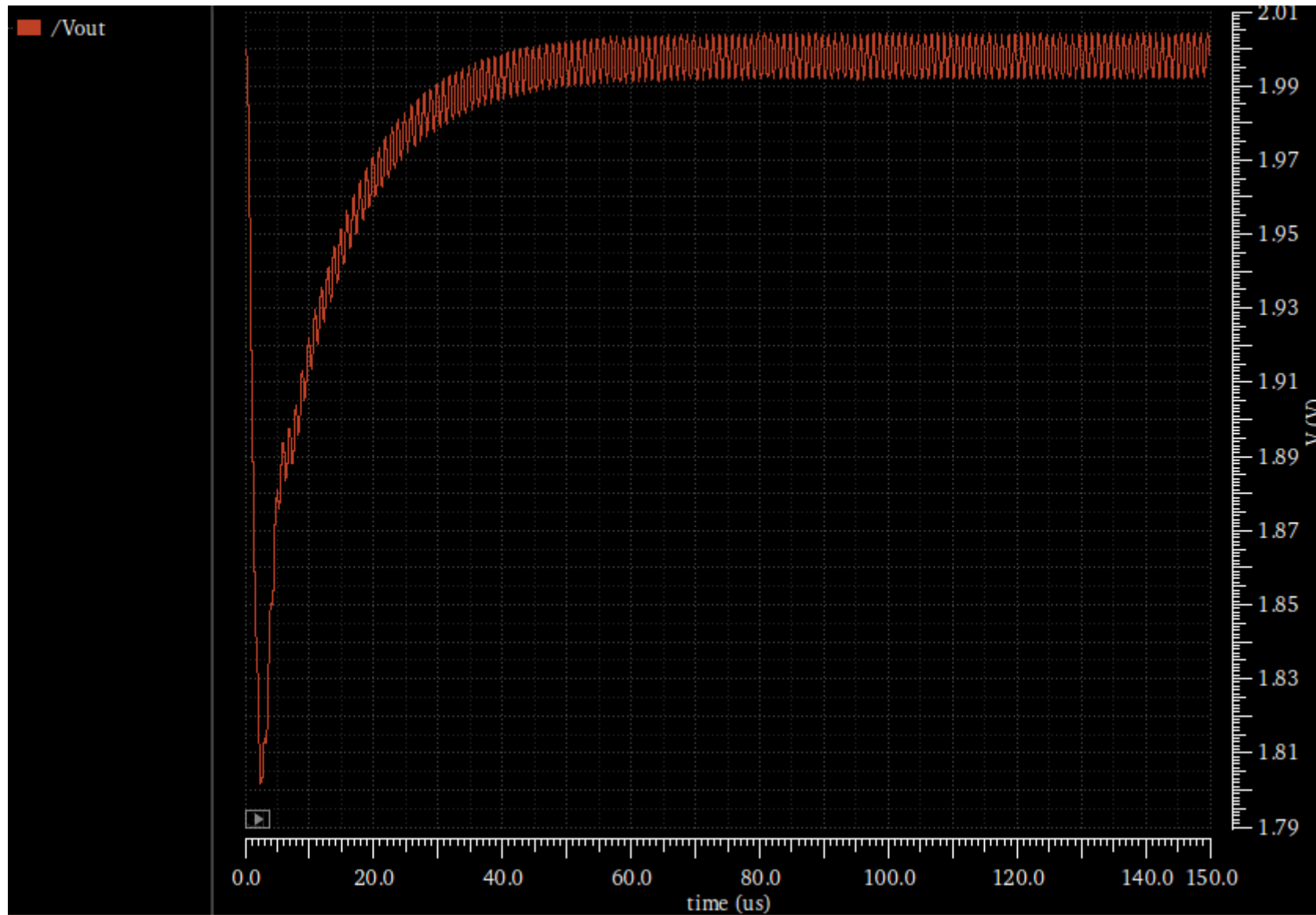
- Go to Tools -> Library Path Editor in CIW
- Edit -> Add Library
- Find the following library
/opt/et4382/Tutorials/Tutorial5
- Click Ok
- File -> Save
- Close the Library Path Editor



**Load the parameters stored in “cellview”.*

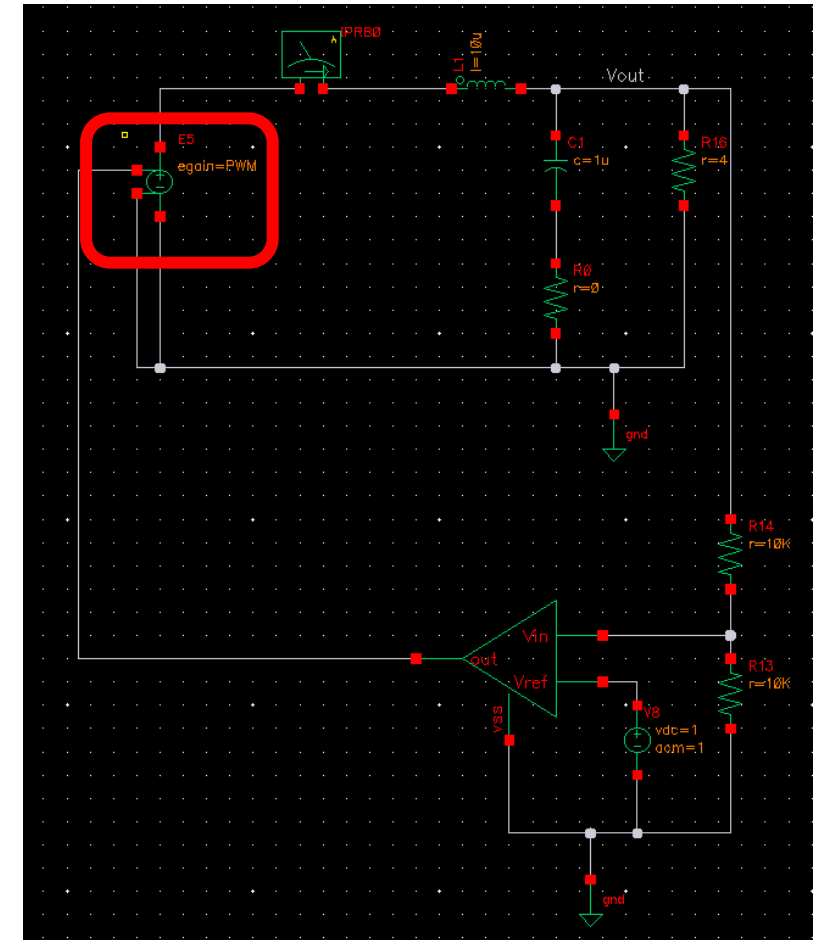
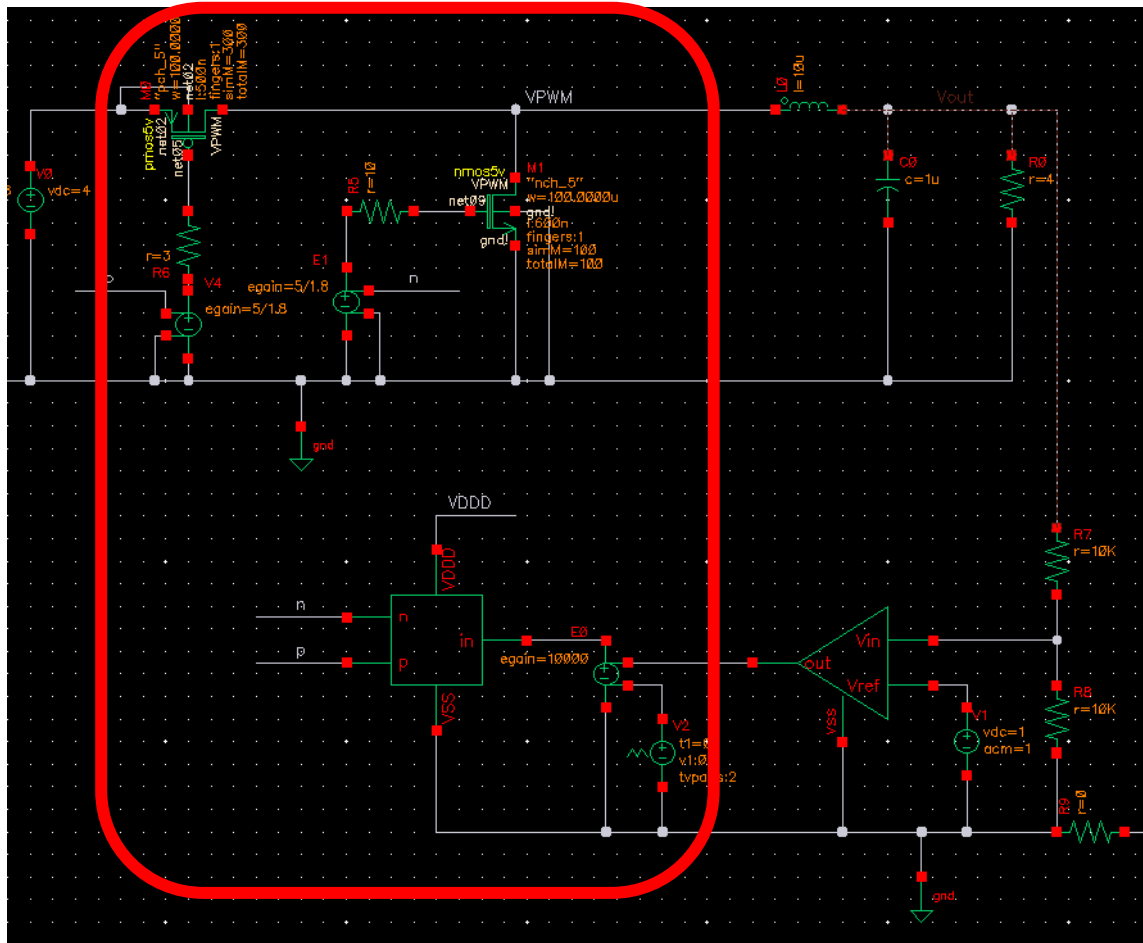


A1-

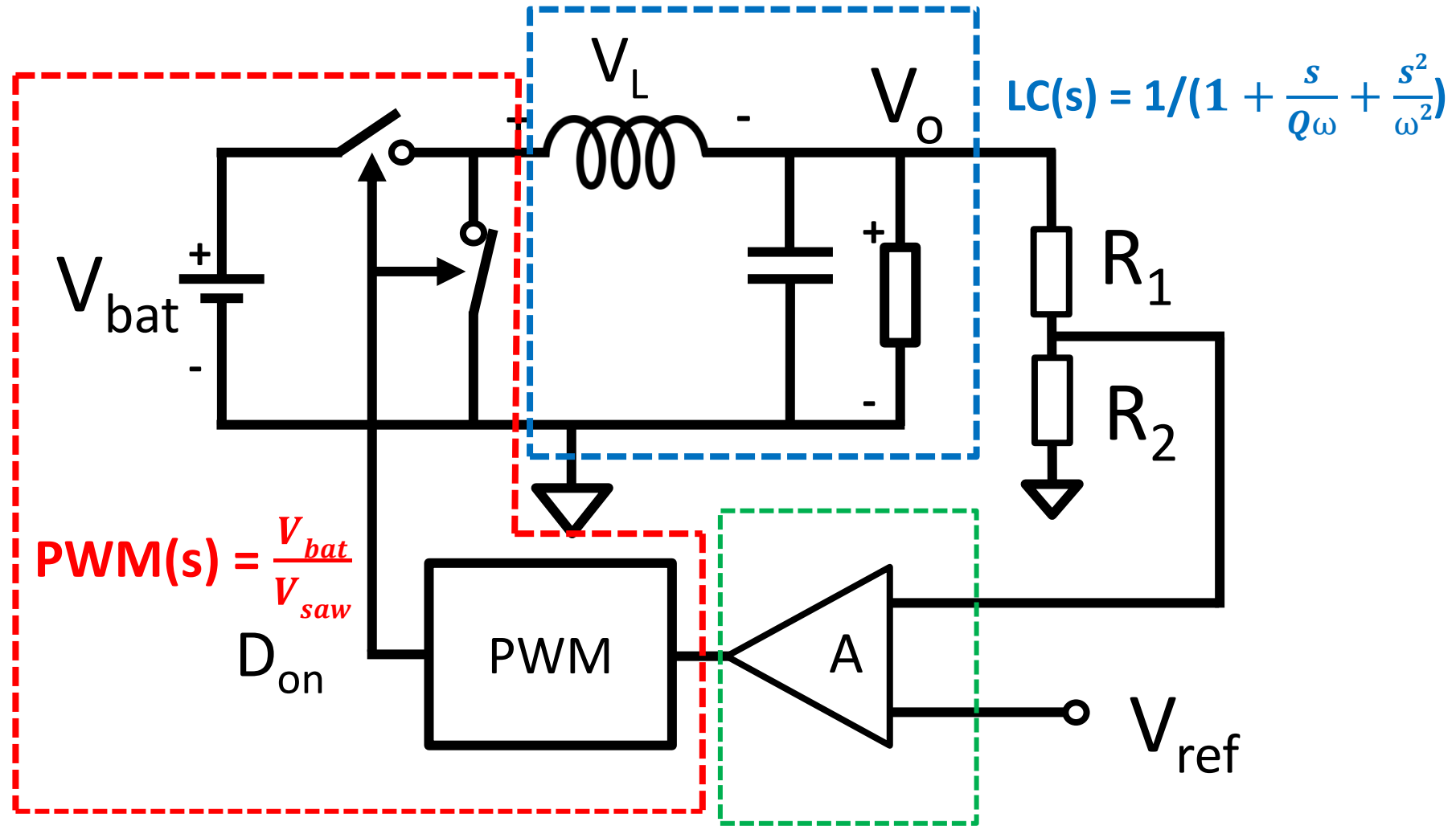


Q2- Open “bucklinear” which is the linearized model of “buckvoltagemode” to run stability analysis. What is the value of variable “PWM” (gain of the VCVS)?

**Load the parameters stored in “cellview”.*



$$A2- PWM = \frac{V_{bat}}{V_{saw}} = \frac{4V}{1V} = 4$$



$$\frac{\Delta D_o}{\Delta V_{in}} = \frac{1}{V_{saw}}, \frac{\overline{\Delta V_{out}}}{\Delta D_o} = V_{bat}$$

Q3- Run stability analysis. Find unity gain frequency, gain margin, and phase margin.

1. Open ADE L and go to Analyses -> Choose

2. Select "stb" analysis and set frequency as a sweep variable.

3. Press "Select" button to choose IPRB from the schematic. Select "IPRB" from the schematic. Probe instance name will be filled out.

4. To plot the waveform: Results -> Direct Plot -> Main Form

5. Select "Add to outputs"

6. Click on "Plot"

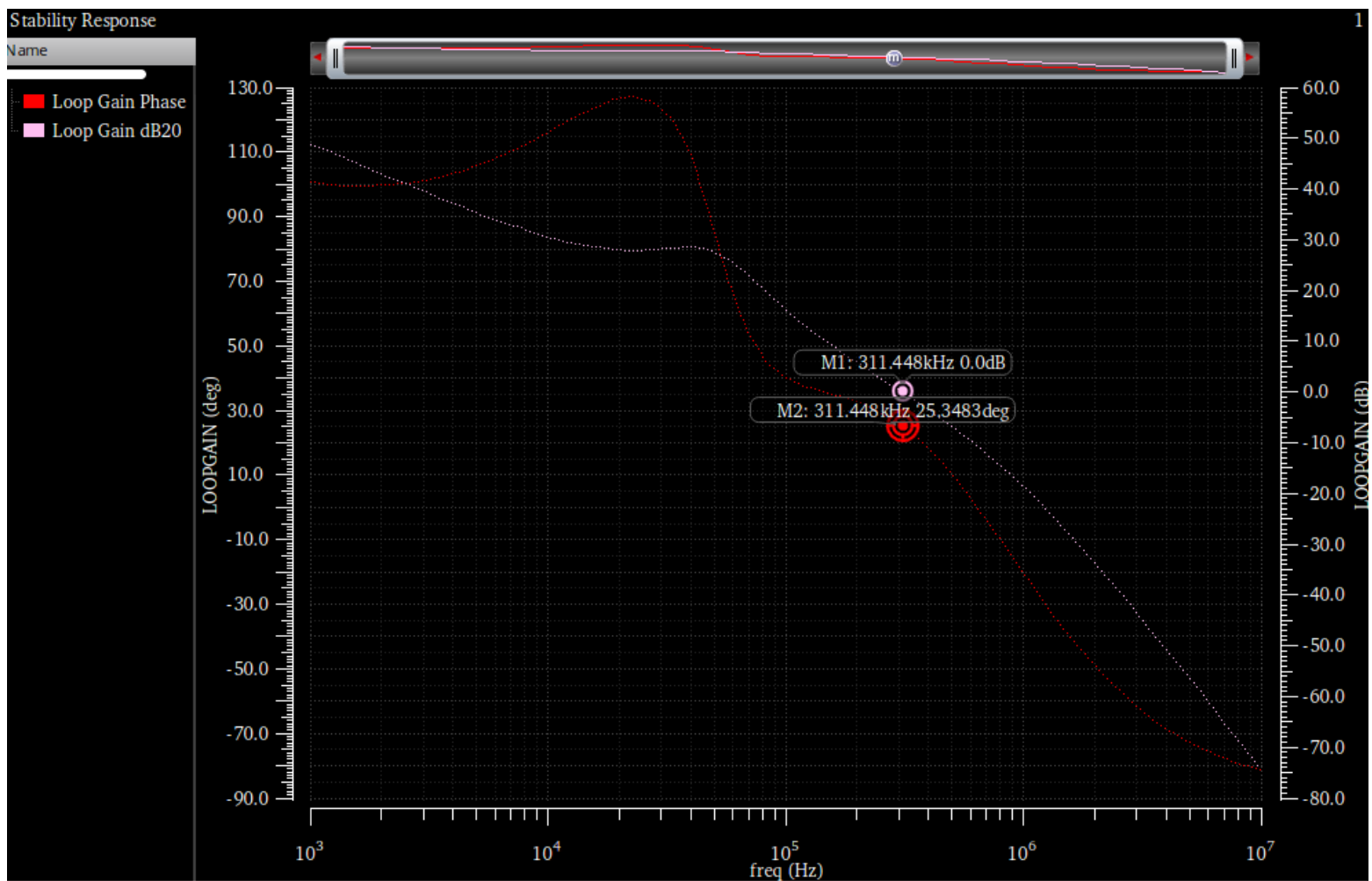
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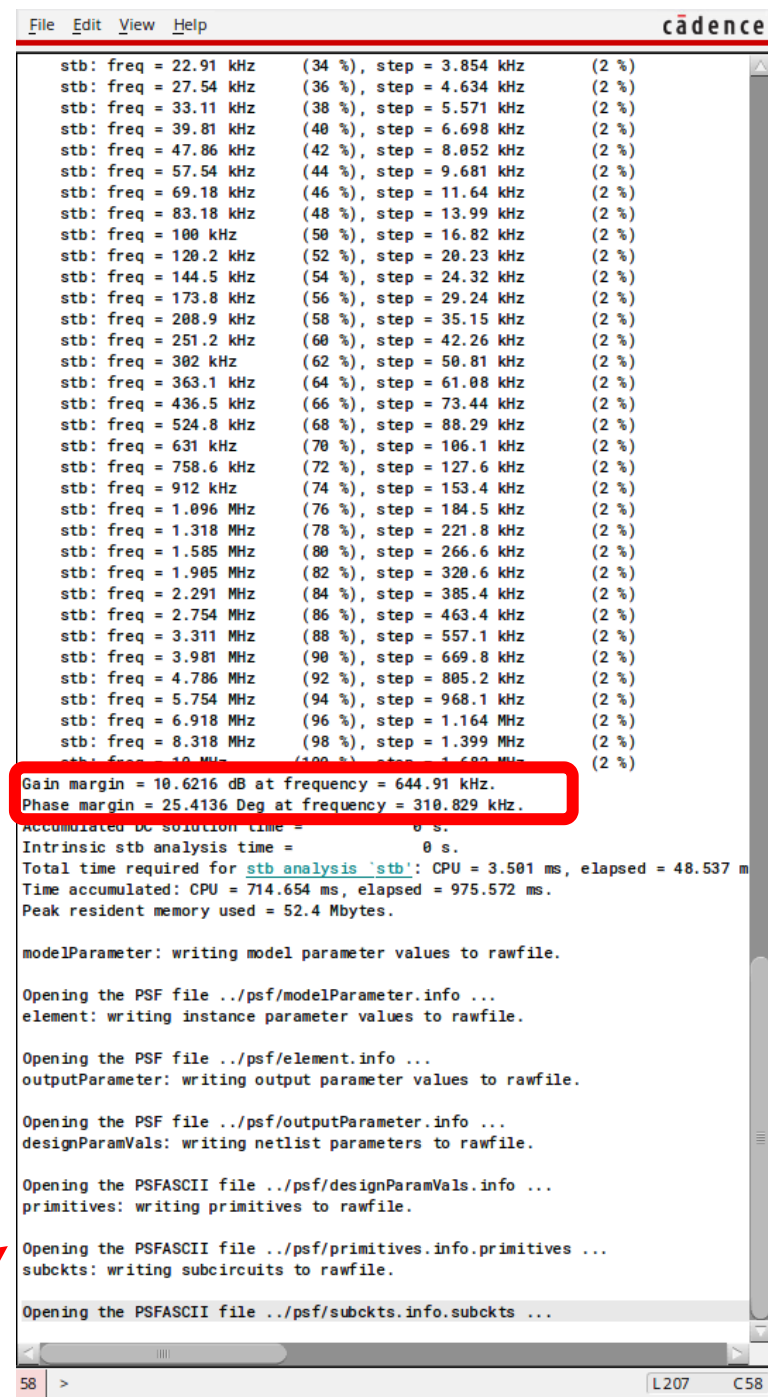
6. Click on "Plot"

A3-



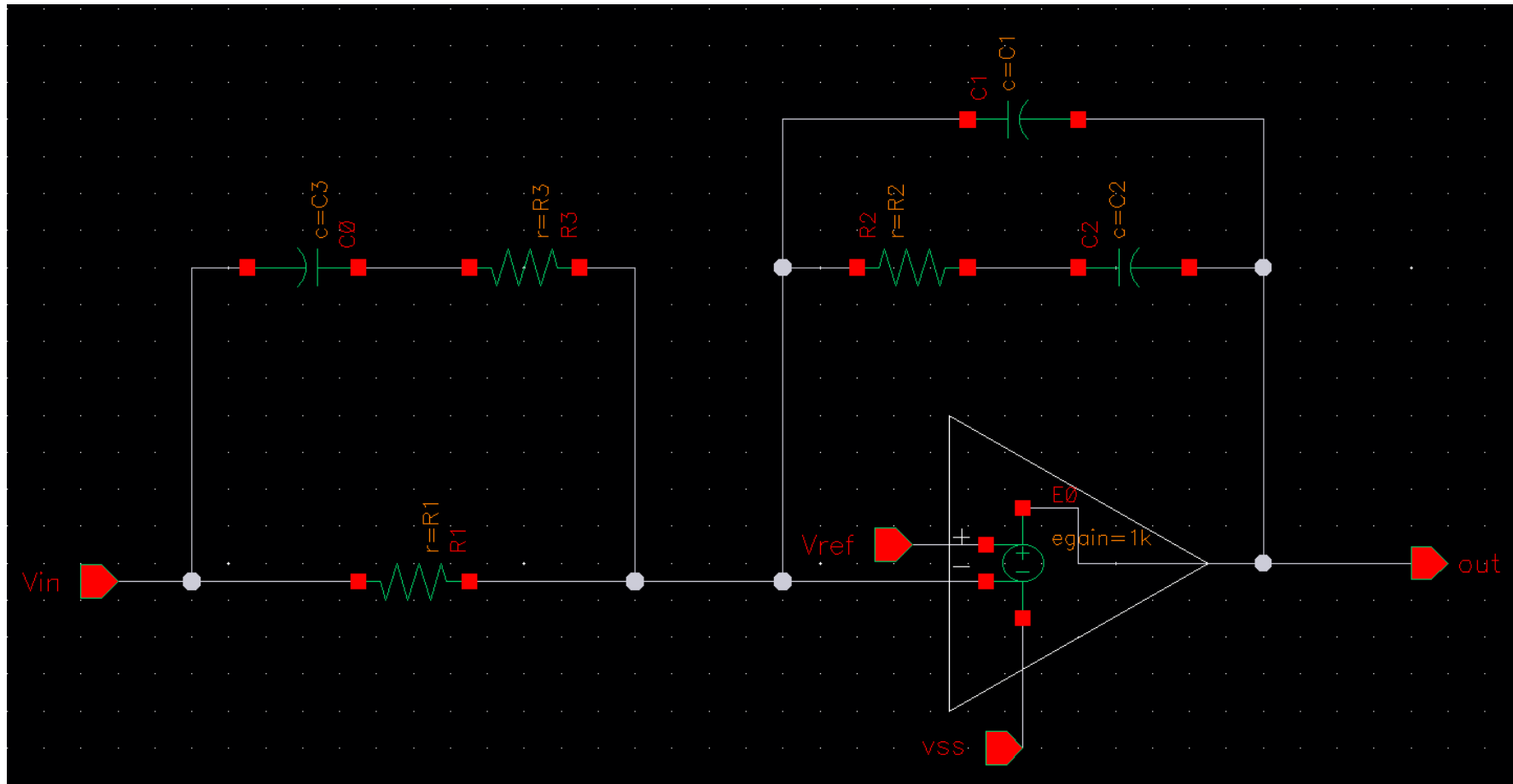
Open log file:

Simulation -> Output Log ...



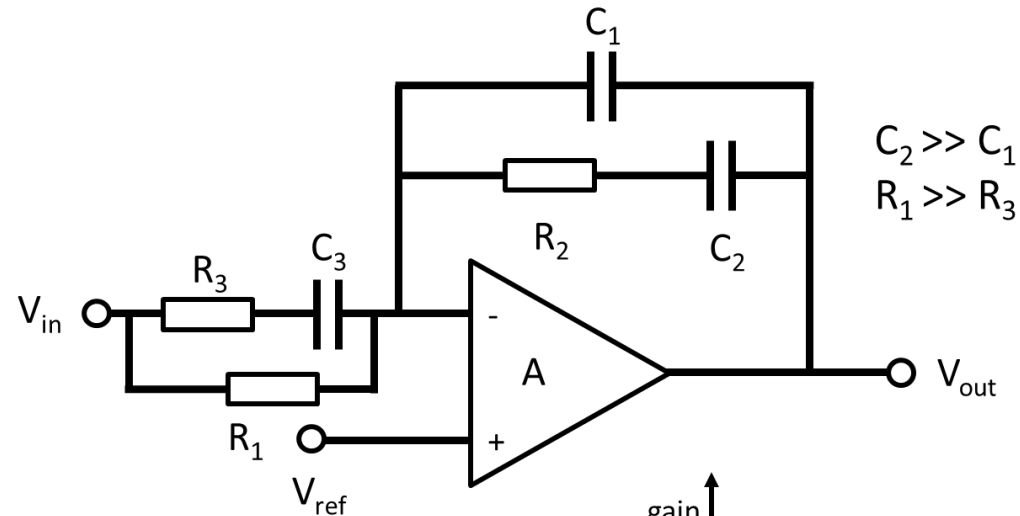
Q4- Open compensator_tb, run ac analysis and plot gain magnitude and phase. What is the mid-band gain of this compensator? Try to decrease it by 6dB.

**Load the parameters stored in cellview.*



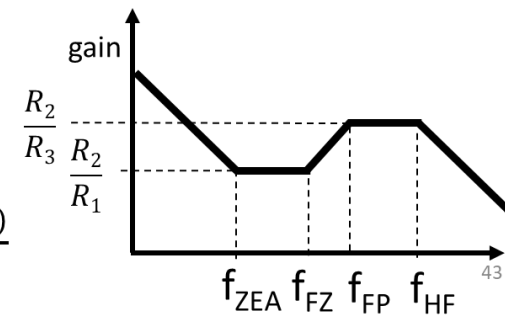
A4- You need to decrease R_2/R_1 to its half. You can decrease R_2 from $500\text{k}\Omega$ to $250\text{k}\Omega$ Notice f_{HZ} also increases to $2\times$

Type III Compensator

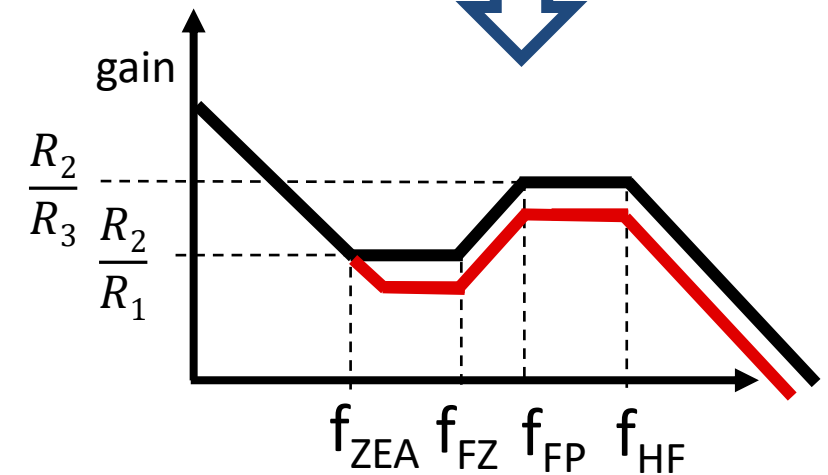
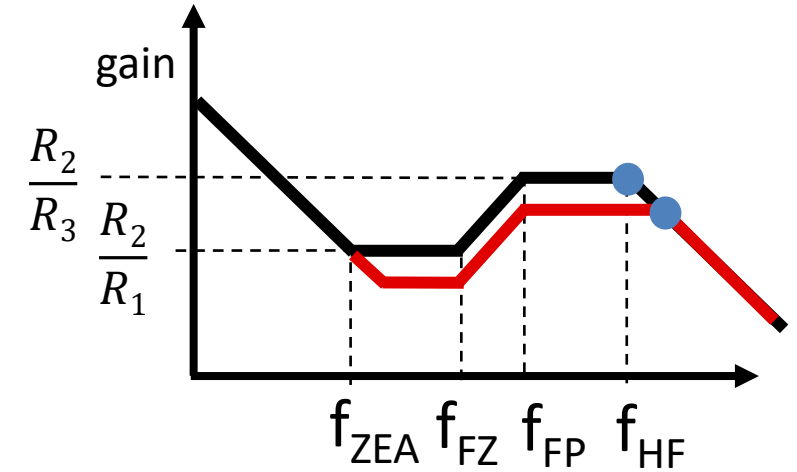
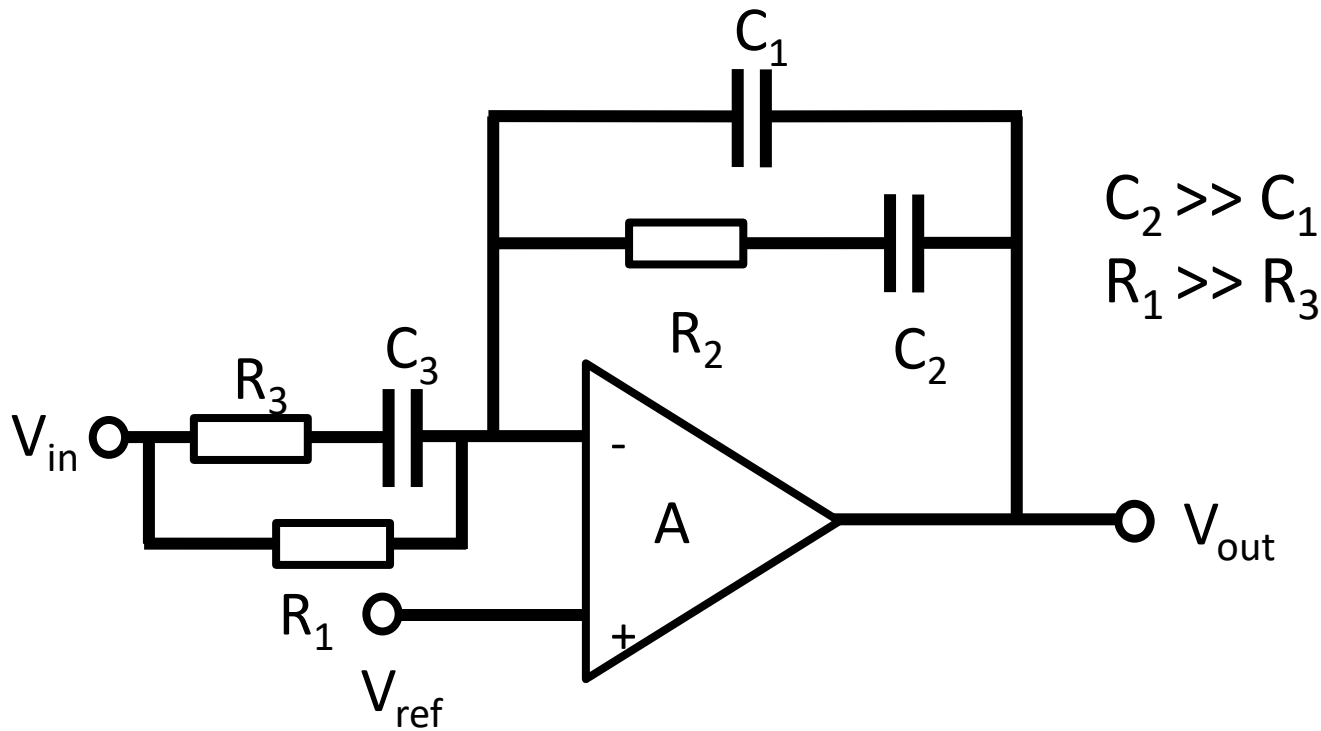


$$H(s) = -A_{VM1} \frac{(1 + \omega_{ZEA}/s)(1 + s/\omega_{FZ})}{(1 + s/\omega_{FP})(1 + s/\omega_{HF})}$$

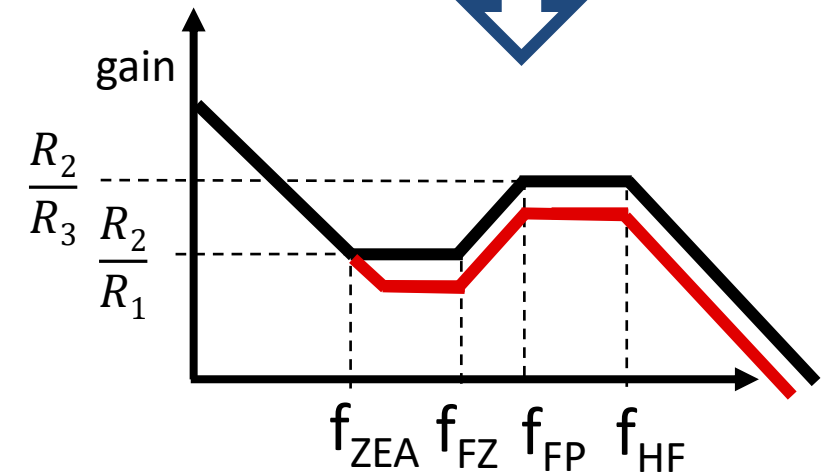
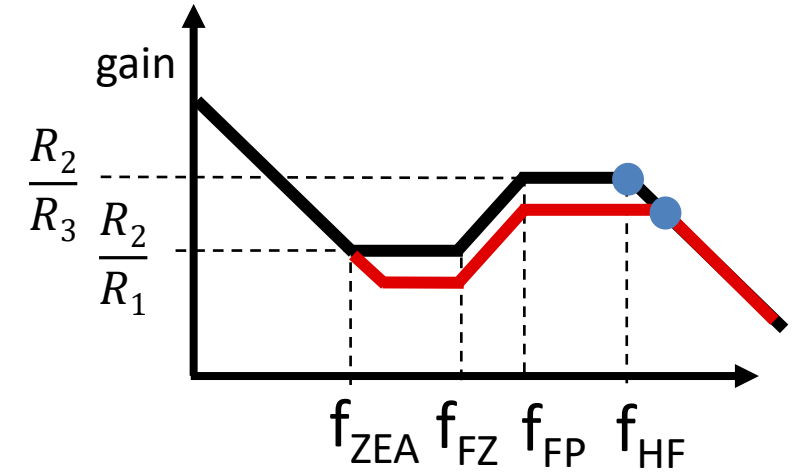
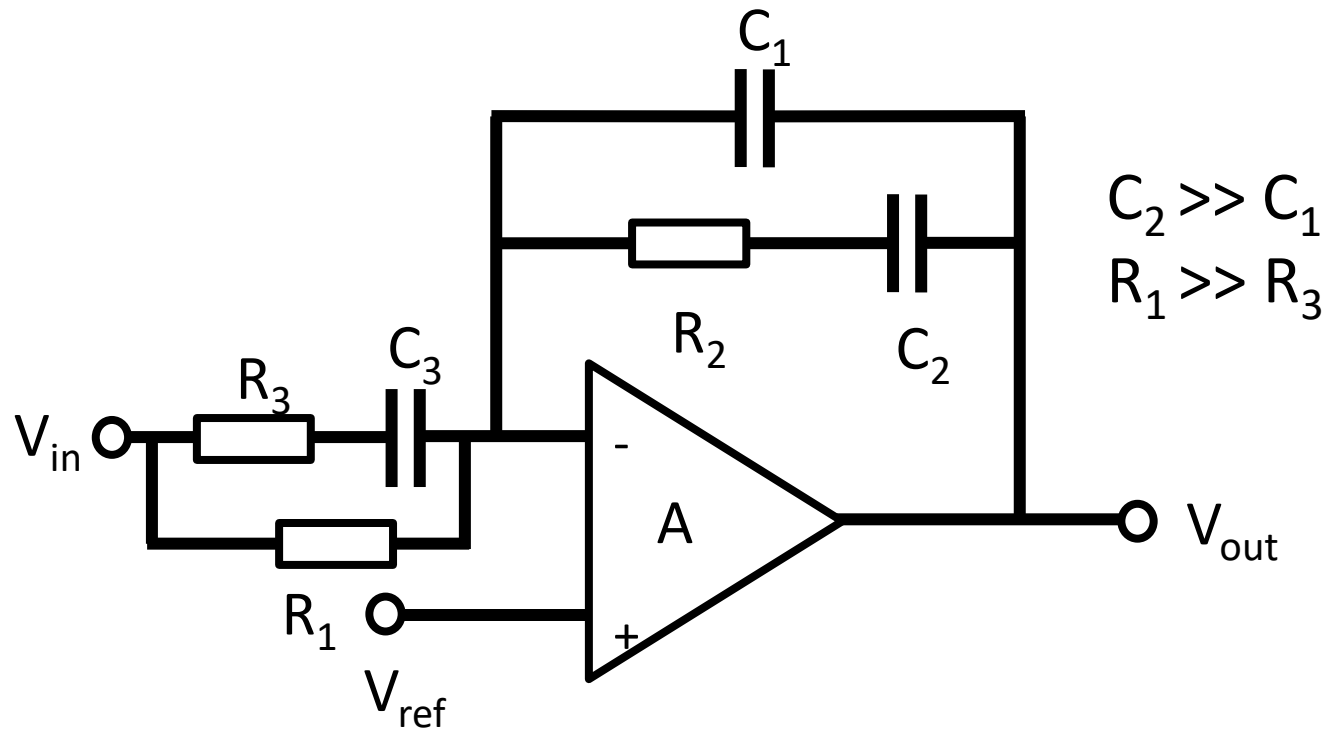
$$= -R_2/R_1 * \frac{\omega_{ZEA}}{s} \frac{(1 + s/\omega_{ZEA})(1 + s/\omega_{ZEA})}{(1 + s/\omega_{FP})(1 + s/\omega_{HF})}$$



Q5- We observed that decreasing R_2 from $500\text{k}\Omega$ to $250\text{k}\Omega$, increases f_{HF} . Can you move f_{HF} to its initial position without changing other poles and zeros?



A5- $f_{HF} = \frac{1}{2\pi R_2 C_1}$, change C_1 from 300fF to 600fF.



Q6- Re-run stb analysis for “bucklinear”, observe how the stability changes with the new compensator parameters?

A6- Phase margin increased from 25° to 28.9° . Only 4 degree improvement.
Can we get better stability?

